

Skills Summary

Differences of Squares and Perfect-Square Trinomials

Use this reference while completing your worksheet or when studying.

Skill 1 — Factoring a difference of two squares

Rule: $a^2 - b^2 = (a - b)(a + b)$

How to spot it: exactly *two* terms, both perfect squares, joined by a *minus* sign. Find a and b by taking the square root of each term: $\sqrt{9x^2} = 3x$, $\sqrt{49} = 7$.

Example: Factor $x^2 - 81$.

Step 1. Two terms, both perfect squares, with a minus between them — that is the difference-of-squares pattern, so no trinomial search is needed.

Step 2. $a = \sqrt{x^2} = x$ and $b = \sqrt{81} = 9$, so the factors are $(a - b)$ and $(a + b)$.

$$x^2 - 81 = (x - 9)(x + 9)$$

Try it: Factor $4x^2 - 25$.

(5 + x2)(5 - x2) :check

(!) **Watch out:** a *sum* of two squares such as $x^2 + 16$ does *not* factor over the whole numbers. The minus sign is part of the pattern.

Skill 2 — Factoring a perfect-square trinomial

Rules: $a^2 + 2ab + b^2 = (a + b)^2$ $a^2 - 2ab + b^2 = (a - b)^2$

How to spot it: *three* terms; the first and last are perfect squares; and the middle term equals $2ab$. The middle-term test is the one that decides it.

Example: Factor $x^2 - 14x + 49$.

Step 1. Three terms with square ends suggests a perfect square, so name a and b and then test the middle term before committing.

Step 2. $a = x$, $b = 7$, and $2ab = 2 \cdot x \cdot 7 = 14x$, which matches the middle term. The middle sign is negative, so use $(a - b)^2$.

$$x^2 - 14x + 49 = (x - 7)^2$$

Try it: Factor $9x^2 + 30x + 25$.

2(5 + x3) :check

(!) **Watch out:** square ends are not enough. $x^2 + 12x + 36$ is a perfect square because $2 \cdot x \cdot 6 = 12x$; $x^2 + 13x + 36$ is not. Always run the middle-term test.

Skill 3 — Common factor first, then the pattern

Rule: factor out the greatest common factor before looking for any special form, then factor what is left. “Factor completely” means keep going until nothing inside the parentheses factors again.

Example: Factor $8x^2 - 18$ completely.

Step 1. Neither term is a perfect square on its own, but both are even — take the common factor out first and look again.

Step 2. $8x^2 - 18 = 2(4x^2 - 9)$, and inside the parentheses $4x^2 = (2x)^2$, $9 = 3^2$, a difference of squares.

$$8x^2 - 18 = \mathbf{2(2x - 3)(2x + 3)}$$

Try it: Factor $5x^2 - 40x + 80$ completely.

check: $5(x - 4)^2$